

CAPSTONE PROJECT PROPOSAL

PROJECT TITLE : E-Commerce UI Automation Testing Framework Using Playwright.

WEBSITE DETAILS :

Website Name: SauceDemo (Swag Labs)

Website URL: <https://www.saucedemo.com>

Automation Tool: Playwright

Language Used: JavaScript / TypeScript

OBJECTIVE :

The objective of this project is to build a scalable End-to-End UI Automation Testing Framework using Playwright for the SauceDemo e-commerce application. The framework will validate important user workflows such as login, product sorting, cart operations, checkout validation, order calculations, and cross-browser execution.

MODULES INCLUDED

1. Authentication & Security

- Valid login, invalid login, session handling, and error validation

2. Product Catalog & Sorting

- Product sorting, product details, and catalog validation

3. Cart Operations

- Add/remove items and cart badge validation

4. Checkout Validation

- Form validation and error message verification

5. Order Calculations

- Subtotal, tax, and total amount validation

6. Menu Navigation & Sidebar

- Sidebar redirects, reset app state, and logout validation

7. Footer Links & Social Media

- Validation of Twitter/X, Facebook, and LinkedIn links

8. Cross-Browser Testing

○ Execution on Chromium, Firefox, and WebKit

KEY FEATURES :

- Page Object Model (POM)
- Cross-Browser Testing
- Reusable Test Scripts
- HTML Reporting
- Parallel Execution
- Screenshot Capture on Failure

PLANNED TEST CASES :

- Core Test Scenarios:40
- Browser Configurations: 3
- Total Test Cases: 120

CONCLUSION :

This project demonstrates an enterprise-level automation testing framework using Playwright with features like POM architecture, cross-browser testing, reusable components, and automated reporting for efficient UI testing.